

See Problem 31 in Exercises 2.5. Properties (2) and (3) suggest the substitutions that can be used to solve a homogeneous differential equation. Specifically, *either* of the substitutions $y = ux$ or $x = vy$, where u and v are new dependent variables, will reduce a homogeneous equation to a *separable* first-order differential equation. To show this, observe that as a consequence of (2) a homogeneous equation $M(x, y) dx + N(x, y) dy = 0$ can be rewritten as

$$x^\alpha M(1, u) dx + x^\alpha N(1, u) dy = 0 \quad \text{or} \quad M(1, u) dx + N(1, u) dy = 0,$$

where $u = y/x$ or $y = ux$. By substituting the differential $dy = u dx + x du$ into the last equation and gathering terms, we obtain a separable DE in the variables u and x :

$$M(1, u) dx + N(1, u)[u dx + x du] = 0$$

$$[M(1, u) + uN(1, u)] dx + xN(1, u) du = 0$$

or
$$\frac{dx}{x} + \frac{N(1, u) du}{M(1, u) + uN(1, u)} = 0.$$

We hasten to point out that the preceding formula should not be memorized; rather, *the procedure should be worked through each time*. The proof that the substitutions $x = vy$ and $dx = v dy + y dv$ also lead to a separable equation follows in an analogous manner from (3).

EXAMPLE 1 Solving a Homogeneous DE

Solve $(x^2 + y^2) dx + (x^2 - xy) dy = 0$.

SOLUTION Inspection of $M(x, y) = x^2 + y^2$ and $N(x, y) = x^2 - xy$ shows that these coefficients are homogeneous functions of degree 2. If we let $y = ux$, then $dy = u dx + x du$ so that, after substituting, the given equation becomes

$$(x^2 + u^2x^2) dx + (x^2 - ux^2)[u dx + x du] = 0$$

$$x^2(1 + u) dx + x^3(1 - u) du = 0$$

$$\frac{1 - u}{1 + u} du + \frac{dx}{x} = 0$$

$$\left[-1 + \frac{2}{1 + u} \right] du + \frac{dx}{x} = 0. \quad \leftarrow \text{long division}$$

After integration the last line gives

$$-u + 2 \ln |1 + u| + \ln |x| = \ln |c|$$

$$-\frac{y}{x} + 2 \ln \left| 1 + \frac{y}{x} \right| + \ln |x| = \ln |c|. \quad \leftarrow \text{resubstituting } u = y/x$$

Using the properties of logarithms, we can write the preceding solution as

$$\ln \left| \frac{(x + y)^2}{cx} \right| = \frac{y}{x} \quad \text{or} \quad (x + y)^2 = cxe^{y/x}. \quad \equiv$$

Although either of the indicated substitutions can be used for every homogeneous differential equation, in practice we try $x = vy$ whenever the function $M(x, y)$ is simpler than $N(x, y)$. Also it could happen that after using one substitution, we may encounter integrals that are difficult or impossible to evaluate in closed form; switching substitutions may result in an easier problem.

Bernoulli's Equation The differential equation

$$\frac{dy}{dx} + P(x)y = f(x)y^n, \quad (4)$$

where n is any real number, is called **Bernoulli's equation** and is named after the Swiss mathematician **Jacob Bernoulli** (1654–1705). Note that for $n = 0$ and $n = 1$, equation (4) is linear. For $n \neq 0$ and $n \neq 1$, the substitution $u = y^{1-n}$ reduces any equation of form (4) to a linear equation.

EXAMPLE 2 Solving a Bernoulli DE

Solve $x \frac{dy}{dx} + y = x^2 y^2$.

SOLUTION We first rewrite the equation as

$$\frac{dy}{dx} + \frac{1}{x}y = xy^2$$

by dividing by x . With $n = 2$, we next substitute $y = u^{-1}$ and

$$\frac{dy}{dx} = -u^{-2} \frac{du}{dx} \quad \leftarrow \text{Chain Rule}$$

into the given equation and simplify. The result is

$$\frac{du}{dx} - \frac{1}{x}u = -x.$$

The integrating factor for this linear equation on, say, $(0, \infty)$ is

$$e^{-\int dx/x} = e^{-\ln x} = e^{\ln x^{-1}} = x^{-1}.$$

Integrating
$$\frac{d}{dx} [x^{-1}u] = -1$$

gives $x^{-1}u = -x + c$ or $u = -x^2 + cx$. Since $u = y^{-1}$ we have $y = 1/u$, and so a solution of the given equation is $y = 1/(-x^2 + cx)$. $\equiv$

Note that we have not obtained the general solution of the original nonlinear differential equation in Example 2, since $y = 0$ is a singular solution of the equation.

Reduction to Separation of Variables A differential equation of the form

$$\frac{dy}{dx} = f(Ax + By + C) \quad (5)$$

can always be reduced to an equation with separable variables by means of the substitution $u = Ax + By + C$, $B \neq 0$. Example 3 illustrates the technique.

EXAMPLE 3 An Initial-Value Problem

Solve the initial-value problem $\frac{dy}{dx} = (-2x + y)^2 + 7$, $y(0) = 0$.

SOLUTION If we let $u = -2x + y$, then $du/dx = -2 + dy/dx$, and so the differential equation is transformed into

$$\frac{du}{dx} + 2 = u^2 + 7 \quad \text{or} \quad \frac{du}{dx} = u^2 - 9.$$

The last equation is separable. Using partial fractions,

$$\frac{du}{(u-3)(u+3)} = dx \quad \text{or} \quad \frac{1}{6} \left[\frac{1}{u-3} - \frac{1}{u+3} \right] du = dx$$

and integrating, then yields

$$\frac{1}{6} \ln \left| \frac{u-3}{u+3} \right| = x + c_1 \quad \text{or} \quad \frac{u-3}{u+3} = e^{6x+6c_1} = ce^{6x}. \quad \leftarrow \text{replace } e^{6c_1} \text{ by } c$$

Solving the last equation for u and then resubstituting gives the solution

$$u = \frac{3(1 + ce^{6x})}{1 - ce^{6x}} \quad \text{or} \quad y = 2x + \frac{3(1 + ce^{6x})}{1 - ce^{6x}}. \quad (6)$$

Finally, applying the initial condition $y(0) = 0$ to the last equation in (6) gives $c = -1$. With the aid of a graphing utility we have shown in **FIGURE 2.5.1** the graph of the particular solution

$$y = 2x + \frac{3(1 - e^{6x})}{1 + e^{6x}}$$

in blue along with the graphs of some other members of the family solutions (6).

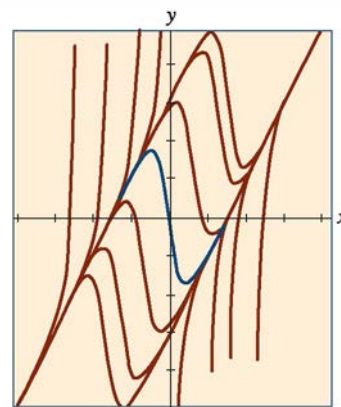


FIGURE 2.5.1 Some solutions of the DE in Example 3

2.5 Exercises

Answers to selected odd-numbered problems begin on page ANS-3.

Each DE in Problems 1–14 is homogeneous.

In Problems 1–10, solve the given differential equation by using an appropriate substitution.

1. $(x - y) dx + x dy = 0$
2. $(x + y) dx + x dy = 0$
3. $x dx + (y - 2x) dy = 0$
4. $y dx = 2(x + y) dy$
5. $(y^2 + yx) dx - x^2 dy = 0$
6. $(y^2 + yx) dx + x^2 dy = 0$
7. $\frac{dy}{dx} = \frac{y - x}{y + x}$
8. $\frac{dy}{dx} = \frac{x + 3y}{3x + y}$
9. $-y dx + (x + \sqrt{xy}) dy = 0$
10. $x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}, x > 0$

In Problems 11–14, solve the given initial-value problem.

11. $xy^2 \frac{dy}{dx} = y^3 - x^3, y(1) = 2$
12. $(x^2 + 2y^2) \frac{dx}{dy} = xy, y(-1) = 1$
13. $(x + ye^{y/x}) dx - xe^{y/x} dy = 0, y(1) = 0$
14. $y dx + x(\ln x - \ln y - 1) dy = 0, y(1) = e$

Each DE in Problems 15–22 is a Bernoulli equation.

In Problems 15–20, solve the given differential equation by using an appropriate substitution.

15. $x \frac{dy}{dx} + y = \frac{1}{y^2}$
16. $\frac{dy}{dx} - y = e^x y^2$
17. $\frac{dy}{dx} = y(xy^3 - 1)$
18. $x \frac{dy}{dx} - (1 + x)y = xy^2$
19. $t^2 \frac{dy}{dt} + y^2 = ty$
20. $3(1 + t^2) \frac{dy}{dt} = 2ty(y^3 - 1)$

In Problems 21 and 22, solve the given initial-value problem.

21. $x^2 \frac{dy}{dx} - 2xy = 3y^4, y(1) = \frac{1}{2}$

22. $y^{1/2} \frac{dy}{dx} + y^{3/2} = 1, y(0) = 4$

Each DE in Problems 23–30 is of the form given in (5).

In Problems 23–28, solve the given differential equation by using an appropriate substitution.

23. $\frac{dy}{dx} = (x + y + 1)^2$
24. $\frac{dy}{dx} = \frac{1 - x - y}{x + y}$
25. $\frac{dy}{dx} = \tan^2(x + y)$
26. $\frac{dy}{dx} = \sin(x + y)$
27. $\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$
28. $\frac{dy}{dx} = 1 + e^{y-x+5}$

In Problems 29 and 30, solve the given initial-value problem.

29. $\frac{dy}{dx} = \cos(x + y), y(0) = \pi/4$
30. $\frac{dy}{dx} = \frac{3x + 2y}{3x + 2y + 2}, y(-1) = -1$

Discussion Problems

31. Explain why it is always possible to express any homogeneous differential equation $M(x, y) dx + N(x, y) dy = 0$ in the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right).$$

You might start by proving that

$$M(x, y) = x^\alpha M(1, y/x) \quad \text{and} \quad N(x, y) = x^\alpha N(1, y/x).$$

32. Put the homogeneous differential equation

$$(5x^2 - 2y^2) dx - xy dy = 0$$

into the form given in Problem 31.

33. (a) Determine two singular solutions of the DE in Problem 10.
 (b) If the initial condition $y(5) = 0$ is as prescribed in Problem 10, then what is the largest interval I over which the solution is defined? Use a graphing utility to plot the solution curve for the IVP.
34. In Example 3, the solution $y(x)$ becomes unbounded as $x \rightarrow \pm\infty$. Nevertheless $y(x)$ is asymptotic to a curve as $x \rightarrow -\infty$ and to a different curve as $x \rightarrow \infty$. Find the equations of these curves.
35. The differential equation

$$\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2$$

is known as **Riccati's equation**.

- (a) A Riccati equation can be solved by a succession of two substitutions *provided* we know a particular solution y_1 of the equation. Show that the substitution $y = y_1 + u$ reduces Riccati's equation to a Bernoulli equation (4) with $n = 2$. The Bernoulli equation can then be reduced to a linear equation by the substitution $w = u^{-1}$.

- (b) Find a one-parameter family of solutions for the differential equation

$$\frac{dy}{dx} = -\frac{4}{x^2} - \frac{1}{x}y + y^2,$$

where $y_1 = 2/x$ is a known solution of the equation.

36. Devise an appropriate substitution to solve

$$xy' = y \ln(xy).$$

≡ Mathematical Model

37. **Population Growth** In the study of population dynamics one of the most famous models for a growing but bounded population is the **logistic equation**

$$\frac{dP}{dt} = P(a - bP),$$

where a and b are positive constants. Although we will come back to this equation and solve it by an alternative method in Section 2.8, solve the DE this first time using the fact that it is a Bernoulli equation.

2.6 A Numerical Method

≡ Introduction In Section 2.1 we saw that we could glean *qualitative* information from a first-order DE about its solutions even before we attempted to solve the equation. In Sections 2.2–2.5 we examined first-order DEs *analytically*; that is, we developed procedures for actually obtaining explicit and implicit solutions. But many differential equations possess solutions and yet these solutions cannot be obtained analytically. In this case we “solve” the differential equation *numerically*; this means that the DE is used as the cornerstone of an algorithm for *approximating* the unknown solution. It is common practice to refer to the algorithm as a *numerical method*, the approximate solution as a *numerical solution*, and the graph of a numerical solution as a *numerical solution curve*.

In this section we are going to consider only the simplest of numerical methods. A more extensive treatment of this subject is found in Chapter 6.

□ Using the Tangent Line Let us assume that the first-order initial-value problem

$$y' = f(x, y), \quad y(x_0) = y_0 \quad (1)$$

possesses a solution. One of the simplest techniques for approximating this solution is to use tangent lines. For example, let $y(x)$ denote the unknown solution of the first-order initial-value problem $y' = 0.1\sqrt{y} + 0.4x^2$, $y(2) = 4$. The nonlinear differential equation cannot be solved directly by the methods considered in Sections 2.2, 2.4, and 2.5; nevertheless we can still find approximate numerical values of the unknown $y(x)$. Specifically, suppose we wish to know the